



Abstract Art

Getting Abstraction “Just Right”

Jeremy Clark
Developer Betterer
GitHub: [jeremybytes](#)

Level: Intermediate

Visual Studio **LIVE!**
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Data Platform **LIVE!**
INSIGHTS FOR MANAGING YOUR DATA ESTATE

TECHMENTOR
IN-DEPTH TRAINING FOR IT PROS

Artificial
Intelligence **LIVE!**
AI FOR DEVELOPERS AND DATA SCIENTISTS

Cloud &
Containers **LIVE!**
CLOUD-NATIVE, PAAS & SERVERLESS COMPUTING

Cybersecurity
& Ransomware **LIVE!**
DEFENDING AGAINST RANSOMWARE AND OTHER ATTACKS

Session Survey

- Your feedback is very important to us
- Please take a moment to complete the session survey found in the mobile app
- Use the QR code or search for “Converge360 Events” in your app store
- Find this session on the Agenda tab
- Click “Session Evaluation”
- Thank you!





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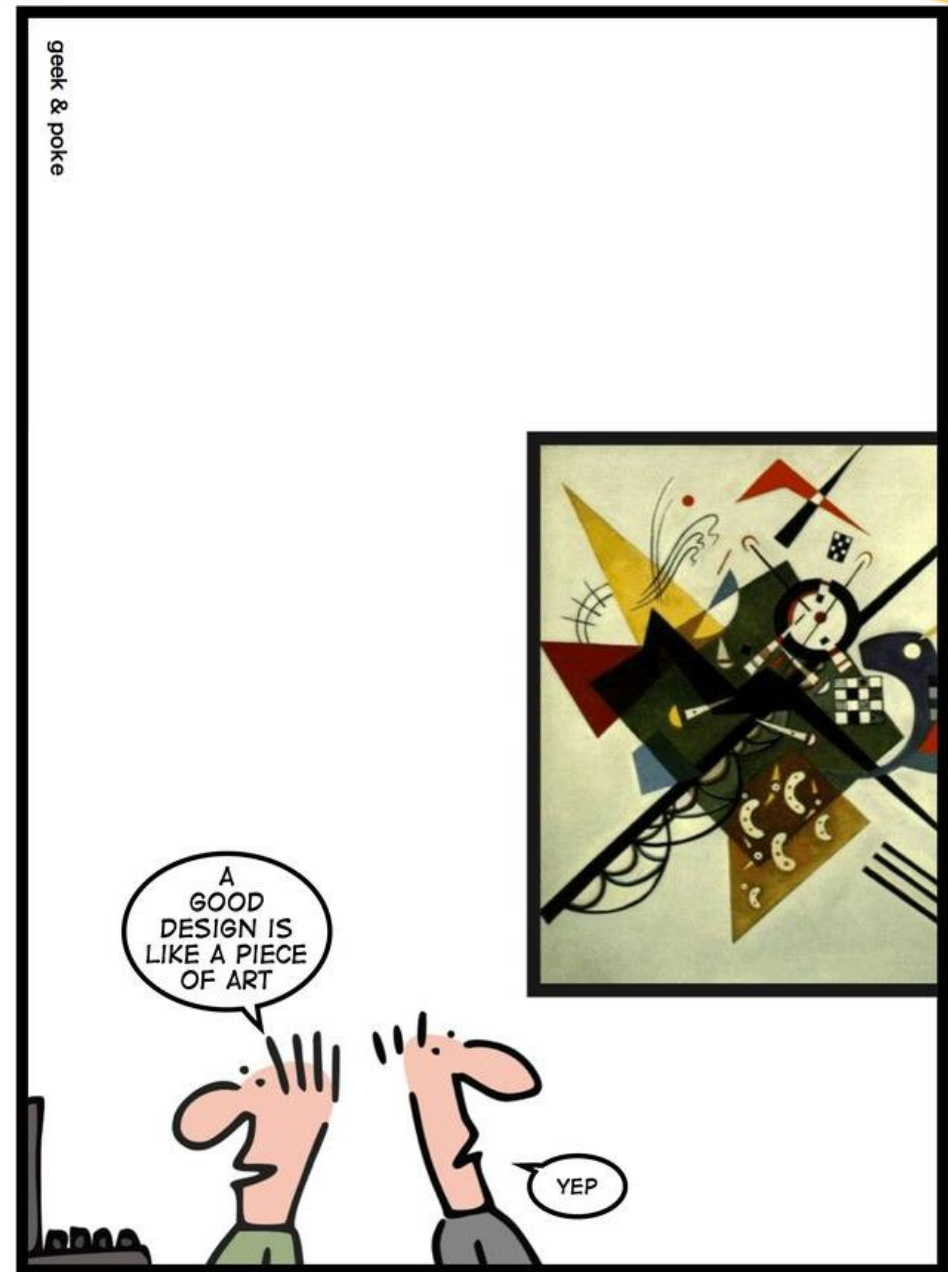
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A Good Design is like A Piece of Art

Geek & Poke – <http://goo.gl/ifu53l>



ABSTRACTION

WHAT IS ABSTRACTION?

Abstraction

- Simplifying reality
- Ignoring extraneous details
- Focusing on what is important for a purpose

ABSTRACTION IS AWESOME!



Maintain



Extend



Test

ABSTRACTION IS AWFUL!



Complexity



Confusion



**Debugging
Difficulty**

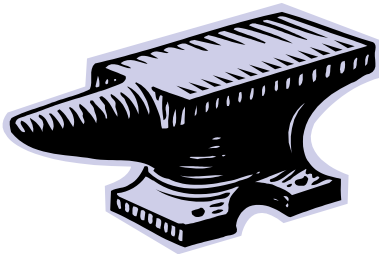
Frustration



https://archive.org/details/goldilocks_and_the_three_bears

GOLDBLOCKS THE DEVELOPER

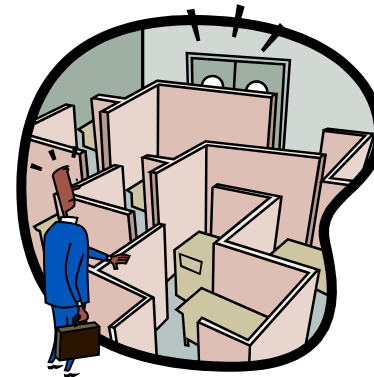
Too Little
Abstraction



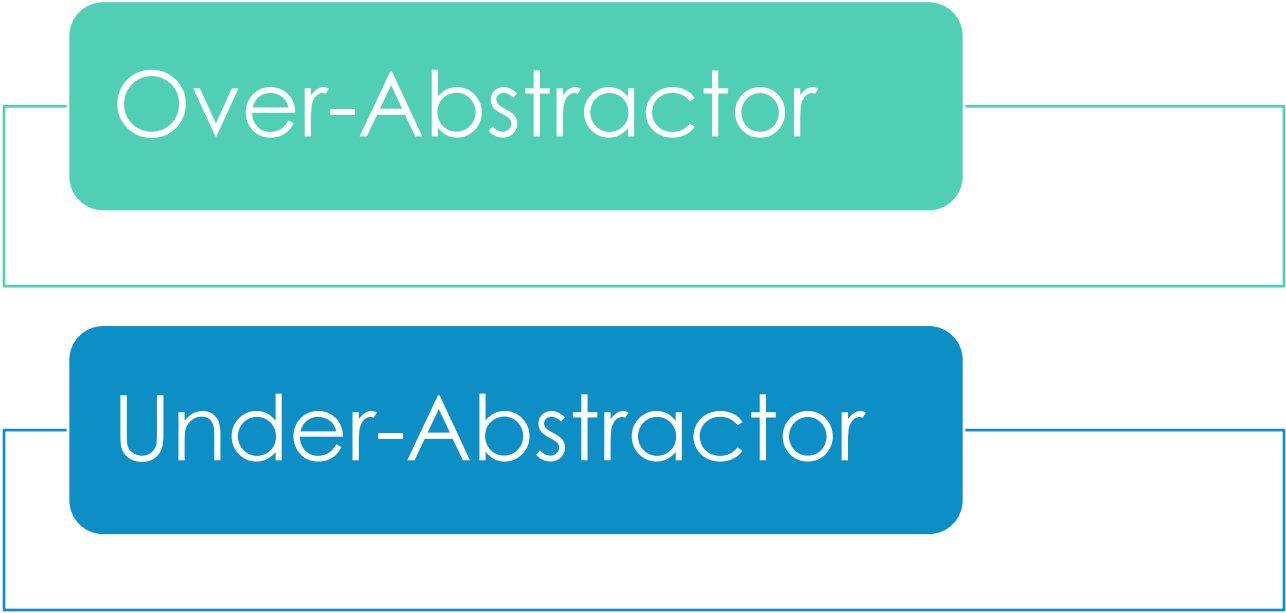
Just Right



Too Much
Abstraction



TWO TYPES OF DEVELOPERS



Over-Abstractor

The diagram consists of two horizontal rows. The top row features a teal rounded rectangle with the text 'Over-Abstractor' in white. To its left and right are thin teal lines that extend to the edges of a larger, empty teal rectangular frame. The bottom row features a blue rounded rectangle with the text 'Under-Abstractor' in white. To its left and right are thin blue lines that extend to the edges of a larger, empty blue rectangular frame.

Under-Abstractor



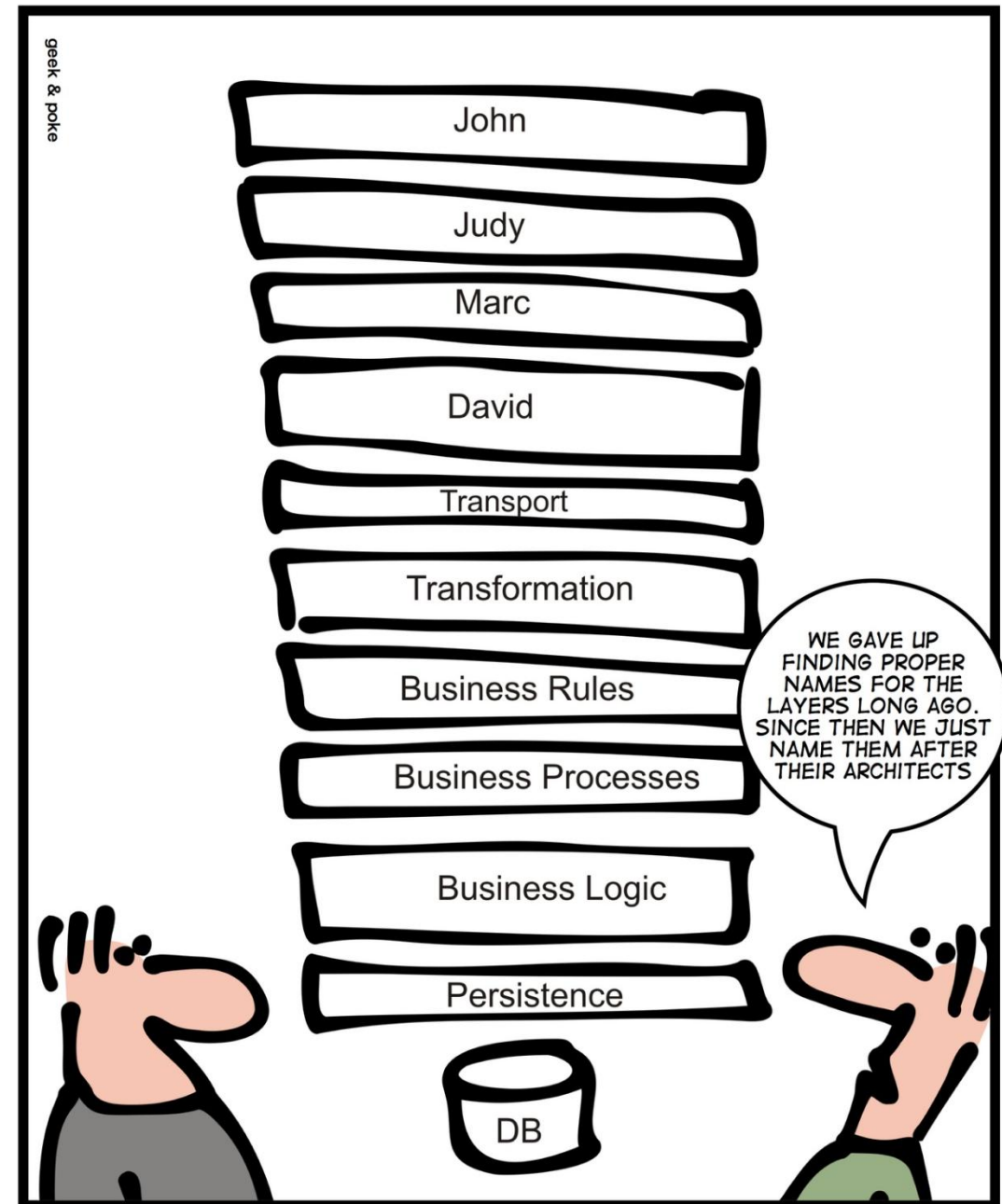
Over-Abstractor

- “We’ll have a good use for this in the future.”
- Overly Complex
- Difficult to Maintain

A Good Architect Leaves A Footprint

Geek & Poke: <http://goo.gl/B4uXa3>

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A GOOD ARCHITECT LEAVES A FOOTPRINT



Under-Abtractor

- “Let’s keep things simple.”
- Rigid
- Difficult to Maintain

COMMON PROBLEM

Over-Abstractor

- “We’ll have a good use for this in the future.”
- Overly Complex
- Difficult to Maintain

Under-Abstractor

- “Let’s keep things simple.”
- Rigid
- Difficult to Maintain



The Default State Quiz

Who Are You?

Let's build a plug-in architecture...

Awesome!
Let's do it.

Maybe we
should look at
compile-time
options.

We need to share a value between modules...

I'll create an object state manager.

Let's use a global variable.

How should we do the UI?

Here's a new
JavaScript
framework.

Let's use the
same framework
we did last time.

Pull data from a database...

ORMs are
awesome!

```
SELECT *  
FROM Customers  
WHERE ID = [@id]
```

We need an object instance...

```
var logger =  
    DIContainer  
    .Resolve<ILogger>()
```

```
var logger =  
    new FileLogger()
```




“
Neither answer is right or wrong. The
correct response is “It depends.”
”

—Jeremy's Standard Response

Let's build a plug-in architecture...

+1

Awesome!
Let's do it.

-1

Maybe we
should look at
compile-time
options.

We need to share a value between modules...

+1

I'll create an object state manager.

-1

Let's use a global variable.

How should we do the UI?

+1

Here's a new
JavaScript
framework.

-1

Let's use the
same framework
we did last time.

Pull data from a database...

+1

ORMs are
awesome!

-1

```
SELECT *  
FROM Customers  
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```

We need an object instance...

+1

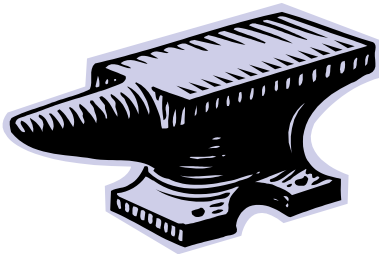
```
var logger =  
    DIContainer  
    .Resolve<ILogger>()
```

-1

```
var logger =  
    new FileLogger()
```

BE HONEST WITH YOURSELF

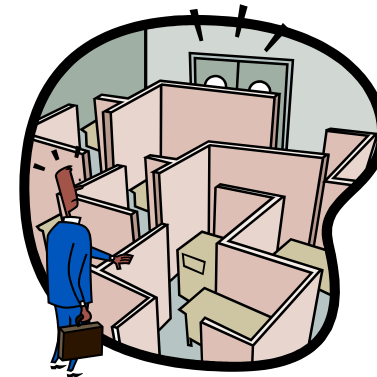
**Too Little
Abstraction**



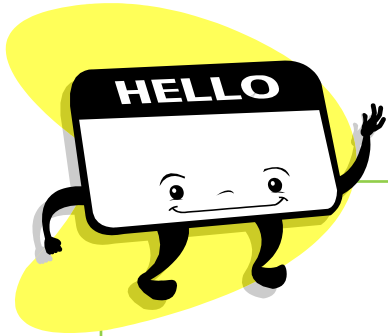
Just Right



**Too Much
Abstraction**



WHO AM I?



Under-Abstractor

- Hello. My name is Jeremy, and I'm an Under-Abstractor.

“Keep Things Obvious”

“Don't Be Tricky”

REPORTING APPLICATION

The image displays a Reporting Application interface with three overlapping windows. The background window is the 'Report Viewer' window, which has a red title bar and contains a sidebar with three buttons: 'Report List', 'Add Report', and 'Update Report', each with a right-pointing arrow icon. The main area of this window shows a table with two columns: 'Report Title' and 'Category'. The table contains the following data:

Report Title	Category
Quarterly Financials	
Weekly Sales	
Inventory	
Pending Shipments	
Incoming Shipments	
Advertising Sales	

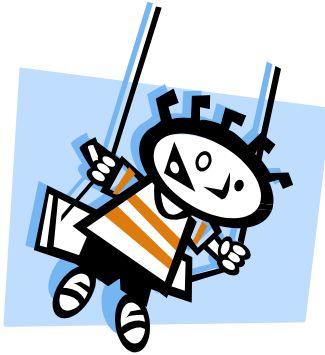
Overlaid on the bottom right is a 'Pending Shipments' window, also with a red title bar. It contains a 'Report List' sidebar and a main area with the following fields:

- Start Date:** 8/24/2014
- End Date:** 8/24/2014
- Additional Info:**
- Calendar:** A calendar for August 2014 showing dates from 27 to 16.

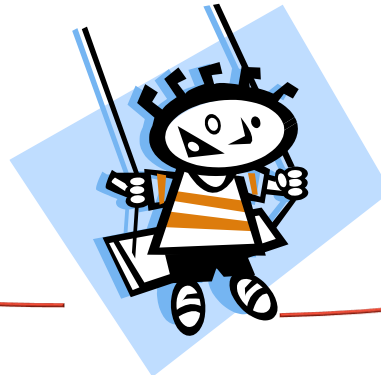
Overlaid on the bottom is a main application window with a light gray title bar. It features a menu bar with 'New Report File', 'Edit Report', and 'Edit Parameters'. Below the menu bar is a tab bar with 'Splash', 'Report List', 'Parameter Values', and 'Report Viewer'. The 'Report Viewer' tab is active, showing a form with 'Report Title' and 'Category' labels and input fields. A 'Run' button is located at the bottom right of this window.

THE PENDULUM EFFECT

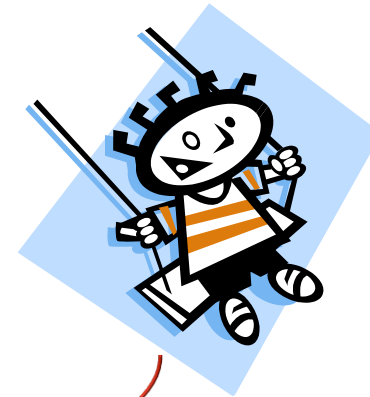
**Under-
Abstraction**



Just Right



**Over-
Abstraction**



REPORTING COMPONENT

[https://reportingServer/RunReport?id=3D2B4067-DBEE-4B3C-A5D7-DC9011300775
&user-id=24¶m1=20240531¶m2=20240606&output=pdf](https://reportingServer/RunReport?id=3D2B4067-DBEE-4B3C-A5D7-DC9011300775&user-id=24¶m1=20240531¶m2=20240606&output=pdf)

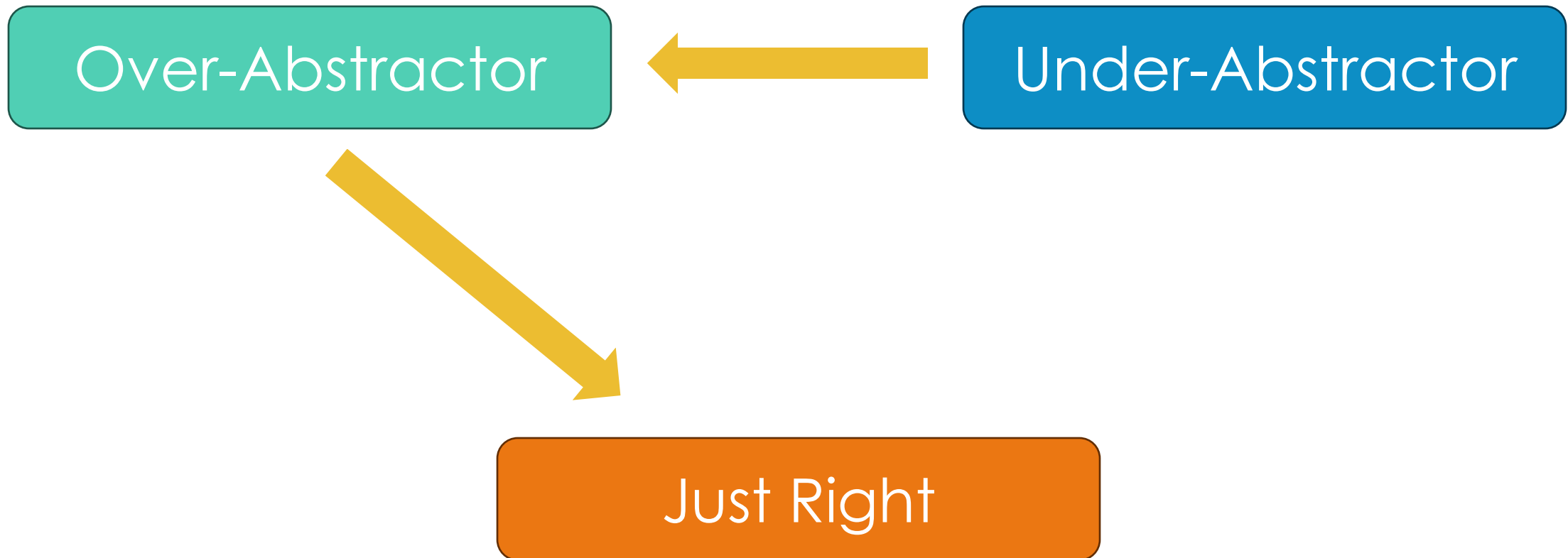
```
var report = new WebReport();  
report.Application = "Facilities";  
report.ReportName = "WeeklyMaintenance";  
report.UserName = userName;  
report.Params.Add("20240531");  
report.Params.Add("20240606");  
report.Output = ReportOutput.PDF;  
  
report.RunReport();
```

THOSE AROUND YOU

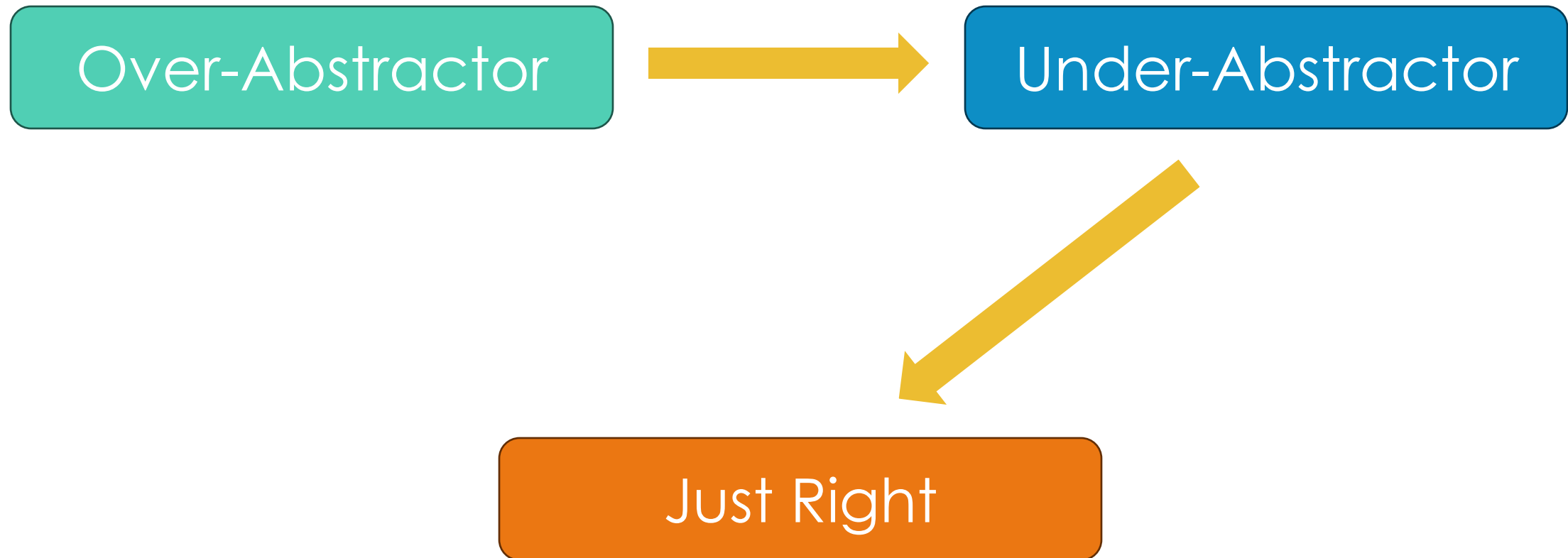
Over-Abstractor

- Jeff loved to build components.
- He liked to create code for re-use.
- He thought of all possible scenarios.

A SYMBIOTIC RELATIONSHIP



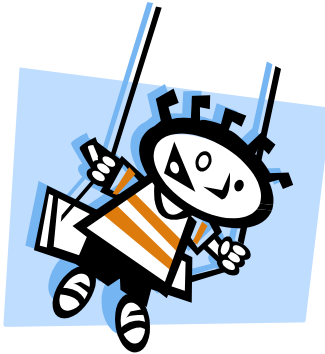
A SYMBIOTIC RELATIONSHIP



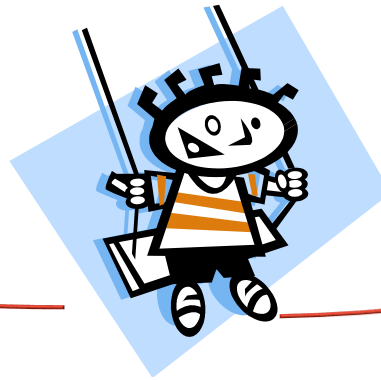


THE PENDULUM EFFECT

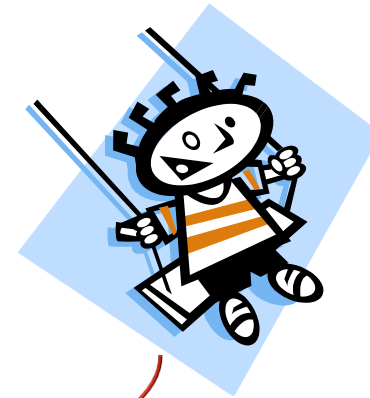
**Under-
Abstraction**



Just Right



**Over-
Abstraction**



“Best” Practices



VARIOUS DATA SOURCES

Microsoft SQL Server

MongoDB

CSV

WebAPI

Oracle

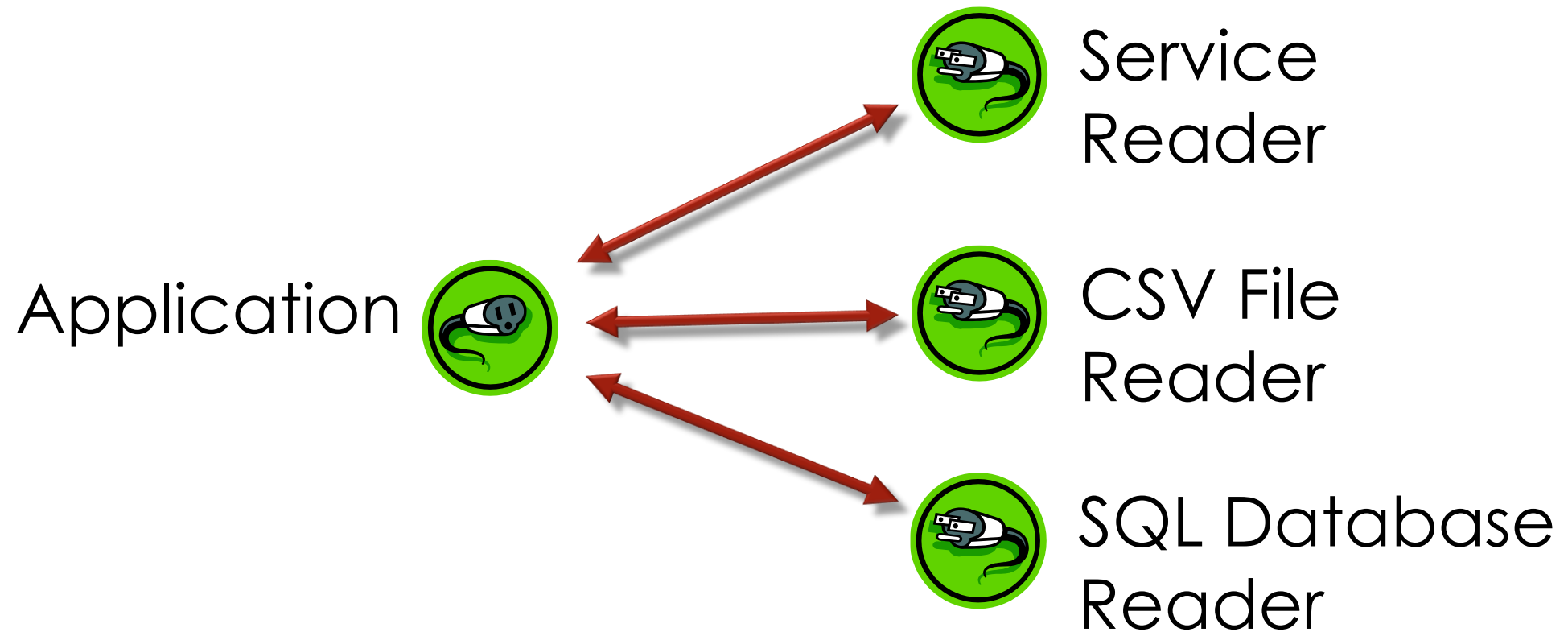
Amazon RDS

JSON

Azure Cosmos DB

Hadoop

PLUGGABLE DATA READERS





DRY

- Don't Repeat Yourself

Under-Abtractor



DON'T REPEAT YOURSELF

Consolidate
Similar
Code

Avoid
Copy/Paste

Copy/Pasta

Spaghetti
Code

SoC

- Separation of Concerns

Under-Abtractor

SINGLE RESPONSIBILITY PRINCIPLE

Complements
Separation of
Concerns

The “S” in
S.O.L.I.D.

A class should
have only one
reason to
change

A class should
do one thing
(and do it well)



YAGNI

- You Ain't Gonna Need It
- (You Aren't Going to Need It)

Over-Abstractor



MORAL OF YAGNI

- Code for the features you have now
- Add abstraction as you need it
- Don't add abstraction based on speculation

We still think about the future,
but we don't implement it yet.



KISS

- Keep It Simple, Stupid
- (Keep It Short & Simple)
- (Keep It Simple & Straightfoward)

Over-Abstractor

DDIY

- Don't Do It Yourself

Over-Abtractor

Under-Abtractor

Over-Abstractor

- Over-Abstractors like to build things to solve specific problems

Under-Abstractor

- Under-Abstractors shy away from external frameworks and libraries

EXAMPLES

Dependency Injection

- Ninject, Autofac, Spring.NET, Microsoft.Extensions.DependencyInjection

Unit Testing Framework

- NUnit, xUnit.net, MSTest, Approval Tests

Mocking

- Moq, NSubstitute, FakeItEasy, JustMock

UI Framework

- Angular, React, Vue, Prism



LEAKY ABSTRACTIONS

All non-trivial abstractions, to some degree, are leaky.

Abstractions fail. Sometimes a little, sometimes a lot. There's leakage. Things go wrong. It happens all over the place when you have abstractions.

--Joel Spolsky

The Law of Leaky Abstractions

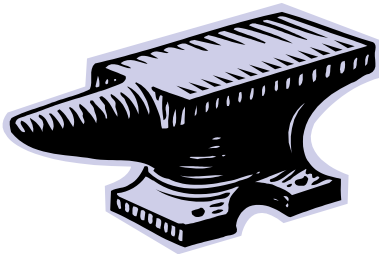
<https://www.joelonsoftware.com/2002/11/11/the-law-of-leaky-abstractions/>

ABSTRACTION IS AWESOME & AWFUL



THE GOLDBLOCKS PRINCIPLE

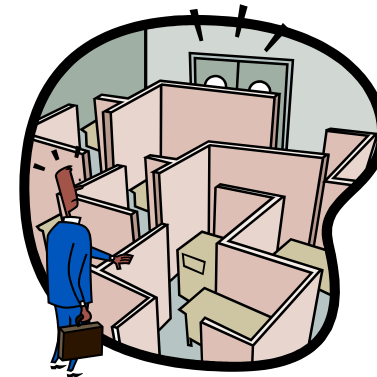
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GETTING THINGS RIGHT

DRY

- Don't Repeat Yourself

SoC

- Separation of Concerns

YAGNI

- You Ain't Gonna Need It

KISS

- Keep It Short & Simple

DDIY

- Don't Do It Yourself

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THANK YOU!

Jeremy Clark

- jeremybytes.com
- youtube.com/jeremybytes
- github.com/jeremybytes

<https://github.com/jeremybytes/vslive2025-orlando>